

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

ANSWERS:

The large difference between the **training MSE (0.85)** and **testing MSE (5.20)** indicates that the neural network performs very well on the training data but poorly on unseen test data. This is a classic sign of **overfitting**.

Analysis of Possible Reasons

1. **Overfitting**
 - The model has memorized the training data instead of learning general patterns.
 - It achieves a low training error but fails to generalize to new data.
2. **Complex Network Architecture**
 - Too many hidden layers or neurons increase the model's capacity, making it prone to fitting noise in the training data.
3. **Insufficient Training Data**
 - A small dataset may not represent all possible patterns, causing poor performance on unseen data.
4. **Noisy or Biased Training Data**
 - Errors, outliers, or irrelevant features in the training dataset can reduce the model's ability to generalize.
5. **Improper Data Splitting**
 - If the training and testing datasets come from different distributions, the test error will be much higher.
6. **Lack of Regularization**
 - Without techniques such as dropout or L1/L2 regularization, the model may become overly dependent on training samples.

Corrective Measures

- **Apply regularization** (L1/L2 regularization, dropout) to reduce overfitting.
- **Use early stopping** to stop training when validation error starts increasing.
- **Simplify the network** by reducing the number of hidden layers or neurons.
- **Increase the size and diversity of the training dataset** or use data augmentation where applicable.
- **Perform proper feature selection** and remove irrelevant or noisy features.
- **Use cross-validation** to evaluate the model's ability to generalize.
- **Ensure proper train-test split** so that both datasets represent the same data distribution.

Conclusion

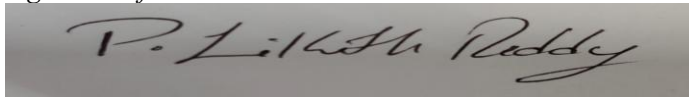
The significant gap between the **training MSE (0.85)** and **testing MSE (5.20)** strongly suggests **overfitting**. The model fits the training data very well but does not generalize effectively to unseen data. Applying regularization, simplifying the model, increasing training data, using early stopping, and validating with cross-validation can significantly improve test performance.

Code of Conduct:

I P LIKITH REDDY (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided